

PriorR 4.2 User Guide

Filtering and prioritization of disease-linked genetic variants

TBLab

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https://www.translationalbioinformaticslab.es · https://github.com/TBLabFJD/PriorR

The screenshots use synthetic demo data (invented trio P001 / P002 / P003), not data of any person.

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1. What is PriorR

PriorR is a filtering and prioritization program of disease-linked genetic variants developed within the Bioinformatics Unit of the Genetics and Genomics Department, IIS-Fundación Jiménez Díaz. It opens variant tables already annotated (SNVs / indels of exomes, genomes or gene panels, and CNVs annotated with AnnotSV) and offers, in a graphical interface, the filters and functions used in the daily analysis: population frequencies, pathogenicity predictors, virtual gene panels, phenotypes (HPO), inheritance patterns and trios, prioritization of variants and genes (GLOWgenes), ACMG criteria, comments and classifications shared by the lab, and the reads of each variant.

> **What is new in PriorR 4.1 and 4.2**
>
> - **Projects**: groups of variant files (one per sample, or files with several samples). Each variant says in how many other samples of the project it is, with a filter by that number and the list of samples (section 15).
> - **Gene structure**: the exons and introns of the transcript with the variant, at scale, and its distance to the borders of the exon or intron (section 12).
> - **Table**: links to the variants that pass the filters and to those *filtered out*; *Var. in gene*; any column can be hidden or moved; sorting keeps the previous order within ties; empty allele frequencies sort as 0; the column filters, search and order are kept when the table changes.

- **Variant information**: the panels with the gene and the other variants of the gene in the sample (parental origin, compound heterozygotes).
- The **patient** in the header; PriorR always opens at `http://127.0.0.1:8484`.
- **CNVs**: the genes and exons taken by the CNV in one line, pass / filtered out, patient in the header and movable columns.

What is new in PriorR 4.0

- Runs on each user's computer (Windows, macOS, Linux) with the data of the lab in a shared folder; no Docker or server needed. Users, groups, comments and classifications shared by the whole lab.
- The same *Open variant file* for exomes, genomes and panels, with **filtering profiles** (None, WES, WGS and the ones saved by the users). Genomes are prefiltered on disk (DuckDB) and load in seconds.
- **Panel library**: PanelApp England and Australia, RetiGene, Orphanet, disease-specific genes of GLOWgenes (Benítez et al. 2026), the lab's panels and HPO terms working as a panel.
- **GLOWgenes ranking** view (precomputed rankings or new rankings run with Docker) and the SGDS specificity score.
- **Trios** and de novo variants; **ACMG** criteria suggested from the annotation; **reanalysis** alerts; **analysis rounds**; filters for every score of the file; AlphaMissense and gnomAD gene constraint.
- Variant information redesigned, external links, **reads viewer** (igv.js) for SNVs and CNVs, in-app Help.

The VCF annotation module of PriorR 3 was removed: the files are annotated by the lab pipeline before PriorR.

2. Requirements and installation

- **Operating system:** Windows, macOS or Linux, 64 bit. **Memory:** 8 GB (16 GB or more for genomes).
- **R** 4.3 or newer (https://cran.r-project.org). The first start needs internet to install the R packages (into the `library/` folder of PriorR, nothing is installed in the system).
- **Docker** only to run new GLOWgenes rankings (optional).

Installers: `PriorR-4.0-setup.exe` (Windows) and `PriorR-4.0.pkg` (macOS) are wizards that install PriorR for the user (no administrator needed) and add a PriorR icon; the Windows one asks for the shared folder of the lab. R must be installed first (the installers say so when it is missing). They are not signed: Windows shows "Windows protected your PC" (*More info* > *Run anyway*) and macOS needs right click > *Open* the first time.

Without installers: copy the PriorR folder to a place every computer can reach (a network folder) or to each computer, and start it with:

System	How to start PriorR
Windows	double click `PriorR.bat`
macOS	double click `PriorR.command`

Linux	`./priorr.sh`

PriorR opens in the web browser (a random local port); closing the console window stops it. It can also run as a server (Dockerfile in the repository) for labs that prefer one shared installation.

3. Configuration of the lab and first start

The lab data (users, groups, comments, classifications, profiles, saved panels, analysis rounds, GLOWgenes rankings) live in a **shared folder**. It is set in `priorr.cfg` next to `launch.R` (configuration of the lab; commented example in `priorr.cfg.example`) or, for one computer, in the configuration of that computer. When nothing says where the shared folder is, the first start shows a short setup wizard that asks for it.

```
shared_folder: //server/genetics/priorr
data_folders: Exomes=//server/exomes; Genomes=//server/genomes
genome_build: GRCh38
memory_limit: 8GB
threads: 4
max_rows: 2000000
```

The first time a shared folder is used, PriorR asks for the first administrator (fig. 1). Afterwards each user signs in once per computer (*Remember me*).

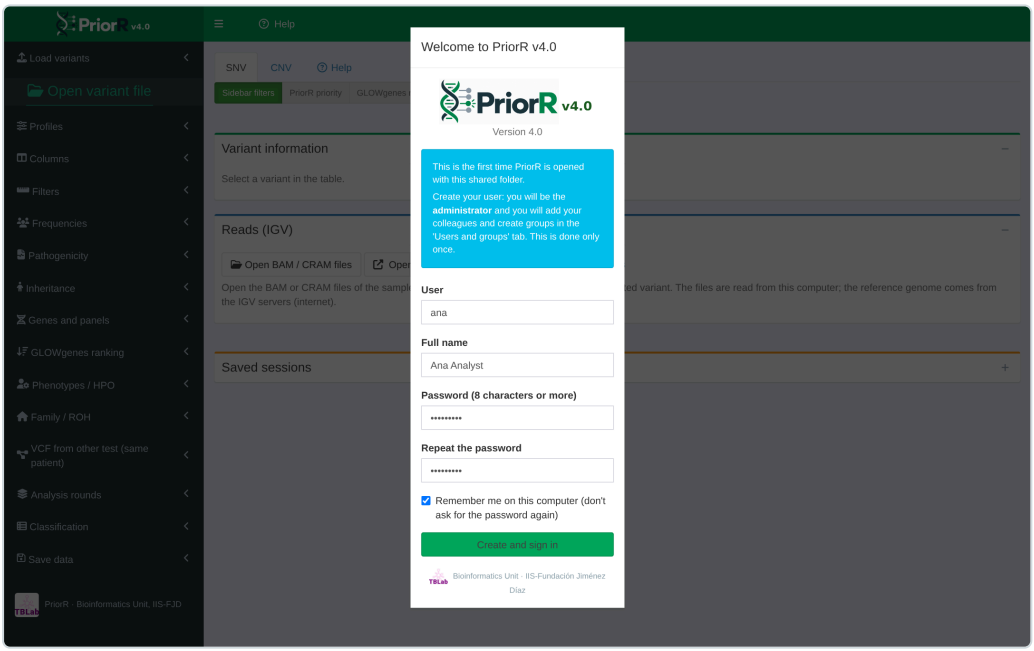


Figure 1. First start with a new shared folder: creation of the first administrator.

4. Input files

- **SNVs / indels:** tab-separated table annotated by the lab pipeline (VEP and others): `CHROM POS REF ALT`, `SYMBOL`, `Consequence`, frequencies (`gnomADg_AF`, `gnomADg_AF_popmax`…), `CLNSIG`, predictors (`CADD_PHRED`, `REVELScore`, `SpliceAI_DS_Max`…) and, for each sample, `<sample>_GT`, `_DP`, `_AD`, `_VAF`. Plain or `.gz`; genomes of several GB are fine. Filters only apply to the columns the file has.

- **CNVs:** AnnotSV tables (`.tsv`).
- **Optional:** PED file of the family, BED of regions, VCF of another test of the same patient, gene lists, BAM / CRAM files with their index.

Demo data: `app/Program_files/examples/demo_trio_synthetic.tsv` and `demo_cnv_synthetic.tsv` (synthetic, made by `scripts/make_demo_data.R`).

5. The interface

The header shows the **patient** of the loaded file (the proband or first sample, with the other samples of the file) and the user (password and sign out). The tabs are **SNV**, **CNV**, **Users and groups** (administrators) and **Help**. On the left, the sidebar with the filters and functions of the tab; in the centre, the table, the variant information, the reads and the saved sessions (fig. 2).

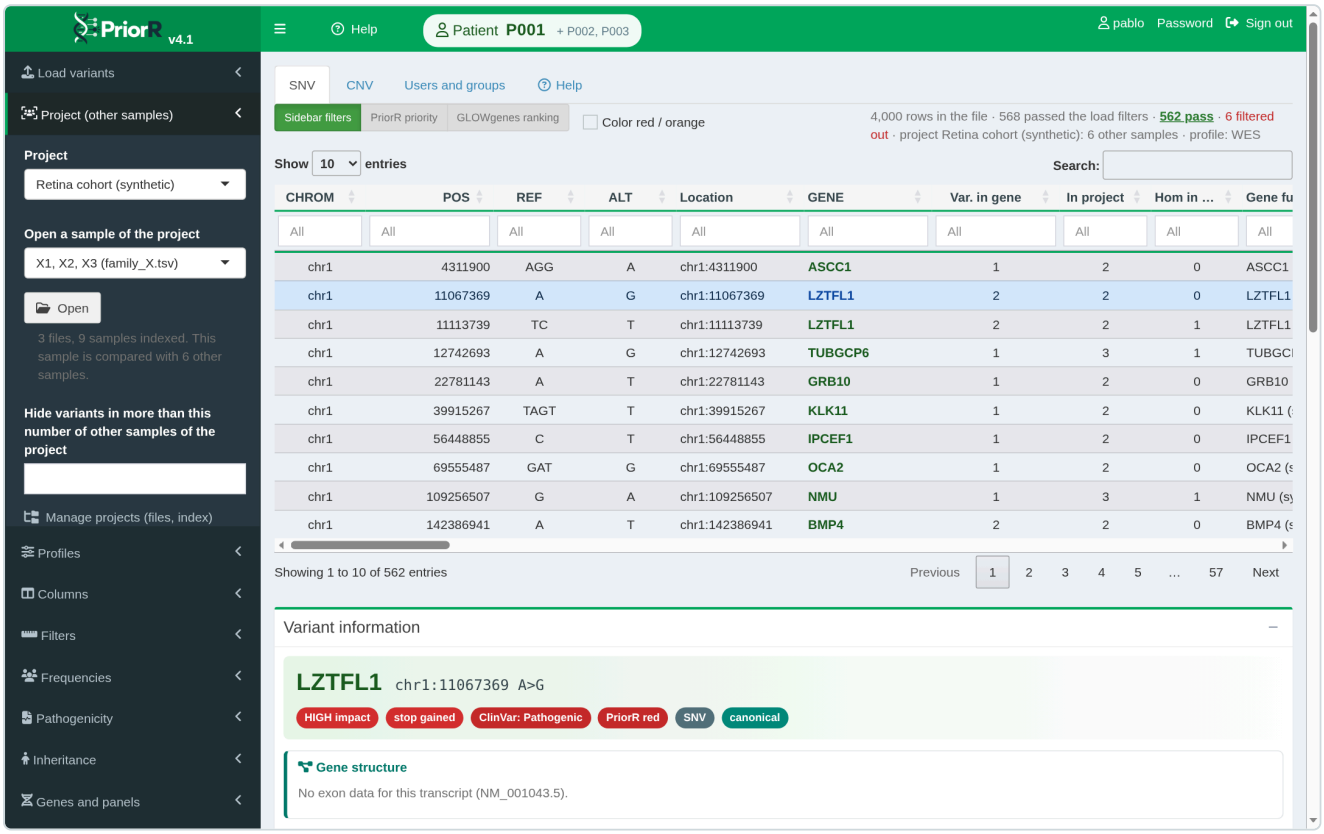


Figure 2. Main interface (SNVs) after loading the demo trio with the WES profile, with a project chosen.

Above the table: the three **views** (*Sidebar filters*, *PriorR priority* and *GLOWgenes ranking*, the one in use in green), the rows of the file and those that passed the load filters, and of them how many **pass** the sidebar filters and how many are **filtered out**: click *filtered out* to see the variants the filters remove (with their details; *Back to … that pass* returns). Column headers show `SYMBOL` as GENE and no underscores; each column has its own filter box; long values are shown in full on hover or click. **Var. in gene** (next to GENE) says how many variants of the gene pass the current filters (compound heterozygotes). Any column can be hidden in *Columns* and moved by dragging its header; the order is kept in profiles and sessions. Clicking a header sorts by it, keeping the previous order within equal values; empty allele frequencies (gnomAD, FJD, CSVS) sort as 0. The column filters, the search and the order are kept when the table changes. *Save data* downloads the table shown.

6. Opening a variant file and filtering profiles

Load variants > Open variant file opens the dialog of fig. 3: choose the file (from the folders of the lab; every file of the folder is listed and a file that is not a variant table gives a

message), a **filtering profile** and, if wanted, genes or panels. The **load filters** decide what is read from the file (essential for genomes); the sidebar filters work afterwards on the loaded variants.

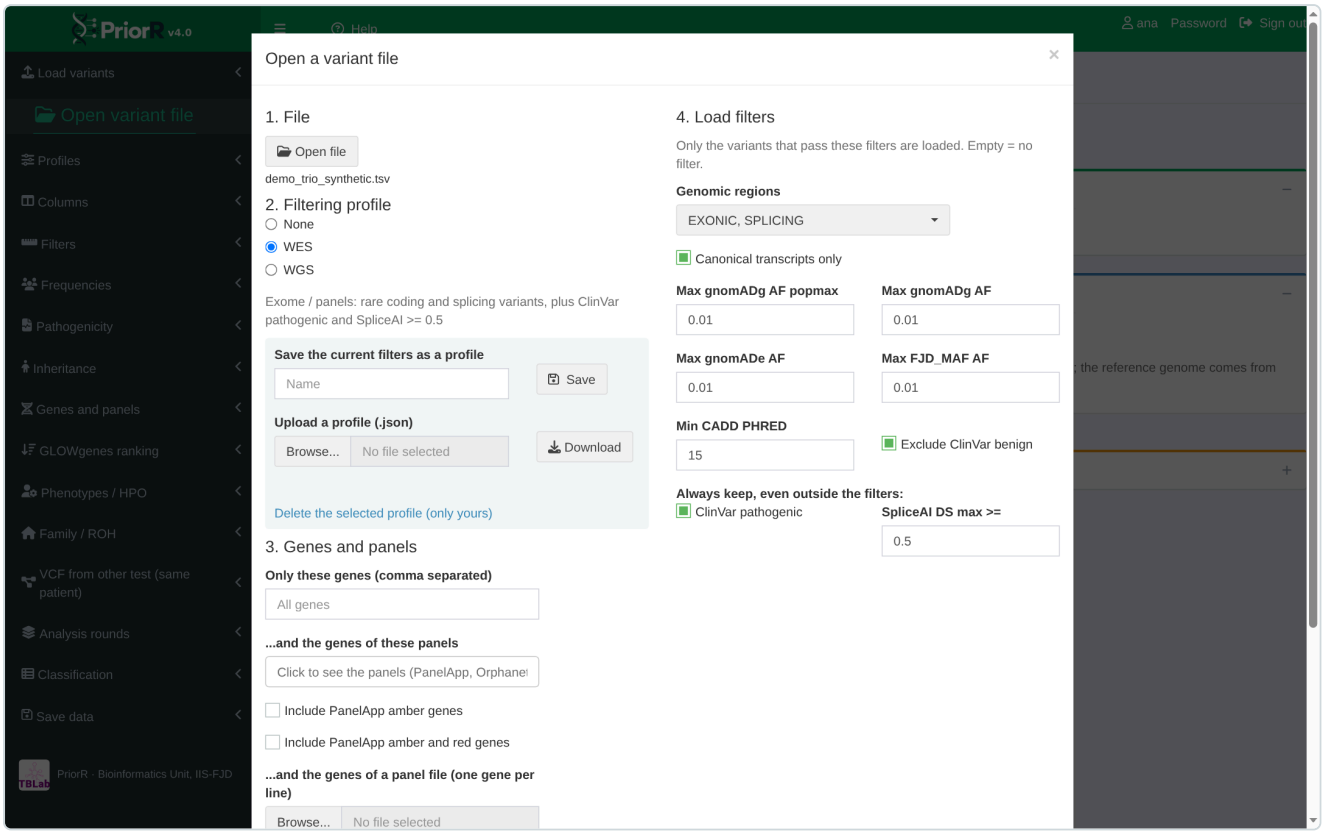


Figure 3. Opening a file with the WES profile: load filters, panels and profile management.

- **None:** every variant. **WES:** rare coding / splicing variants, plus ClinVar pathogenic and SpliceAI ≥ 0.5. **WGS:** the same idea for genomes.
- A profile keeps load filters, sidebar filters and columns. Save the current ones as a profile of your own, upload or download profiles (`.json`); built-in profiles are in `app/Program_files/profiles`.
- Variants rescued by a load rule (ClinVar pathogenic, SpliceAI) can be hidden later by the sidebar filters of the same profile: the line above the table shows loaded vs shown rows.

7. Sidebar filters: frequencies, pathogenicity and scores

- **Columns:** columns shown (hiding columns never changes the result).
- **Filters:** variant type, consequence, genomic region, canonical transcripts, predicted transcripts (XM_ / XR_), BED regions.
- **Frequencies:** maximum gnomAD genomes / popmax / exomes and lab frequency. A missing frequency never removes a variant.
- **Pathogenicity** (fig. 4): minimum CADD and a filter for each score the file has (REVEL, AlphaMissense, MutScore, SpliceAI, dbscSNV ADA / RF, number of pathogenic predictors, DOMINO, phyloP, gene LOEUF). *All must pass* (a missing score passes) or *at least one must pass* (e.g. REVEL ≥ 0.5 **or** SpliceAI ≥ 0.2). Exclude ClinVar benign.

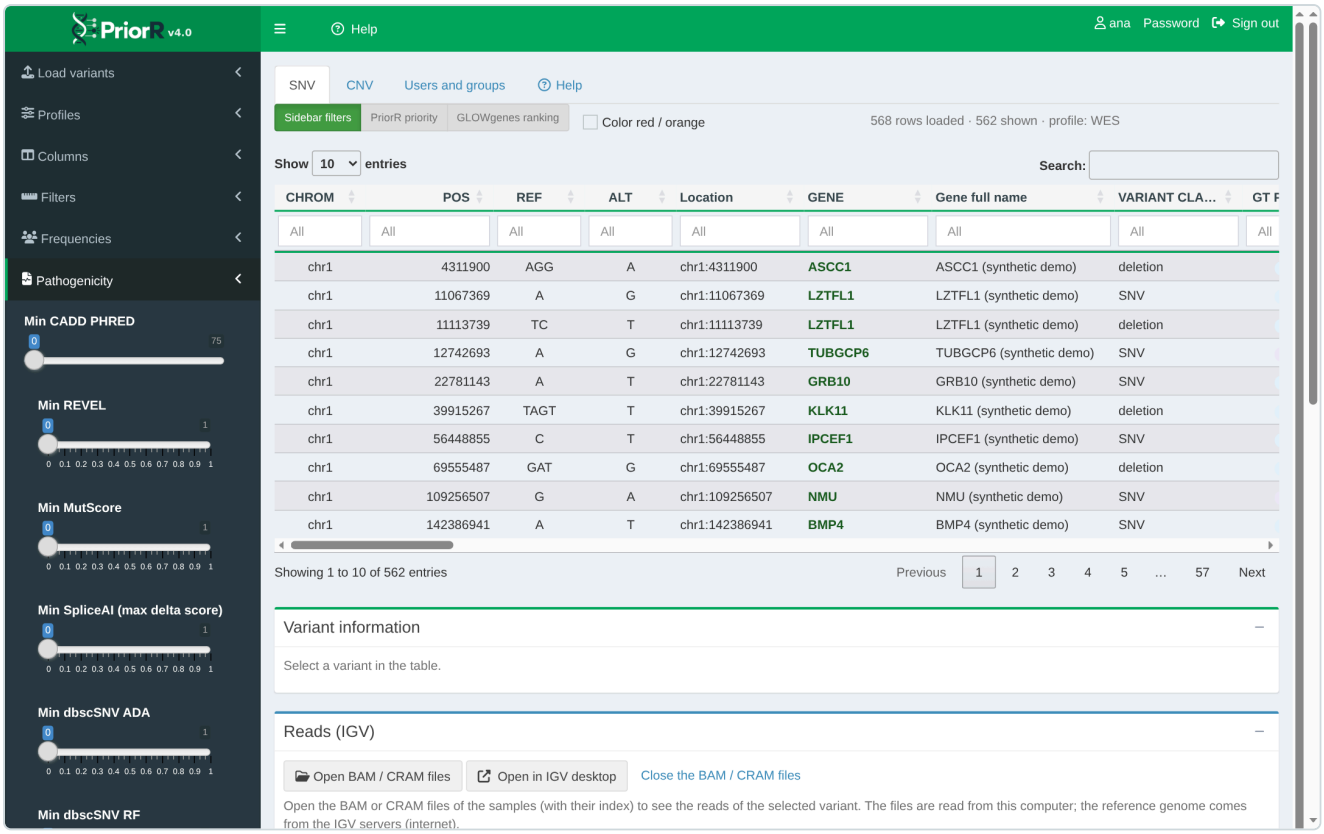


Figure 4. Pathogenicity: only the scores present in the loaded file are offered.

Annotations added by PriorR only when the file doesn't have them: AlphaMissense score and class of missense SNVs (from a file prepared once by the administrator, see section 18) and the gnomAD v4.1 gene constraint (LOEUF, pLI, missense Z). The SGDS column (gene-disease specificity, GLOWgenes) is also added.

8. Genes, panels and phenotypes (HPO)

Genes and panels (fig. 5) keeps the variants of your genes and of the chosen panels:

- **Gene panels:** PanelApp England and Australia (green genes; amber, or amber and red, optionally), RetiGene (inherited retinal diseases), Orphanet and the panels saved by the lab (upload a gene list and save it with a name).
- **Disease-specific genes (Benítez et al., 2026):** genes predicted by GLOWgenes as specific of one disease or of a disease class, in their own list.
- **Exclude** the genes of other panels or of a file.
- **Download the latest PanelApp (only for me):** each user can download the current PanelApp for themselves; the others keep the lab copy.

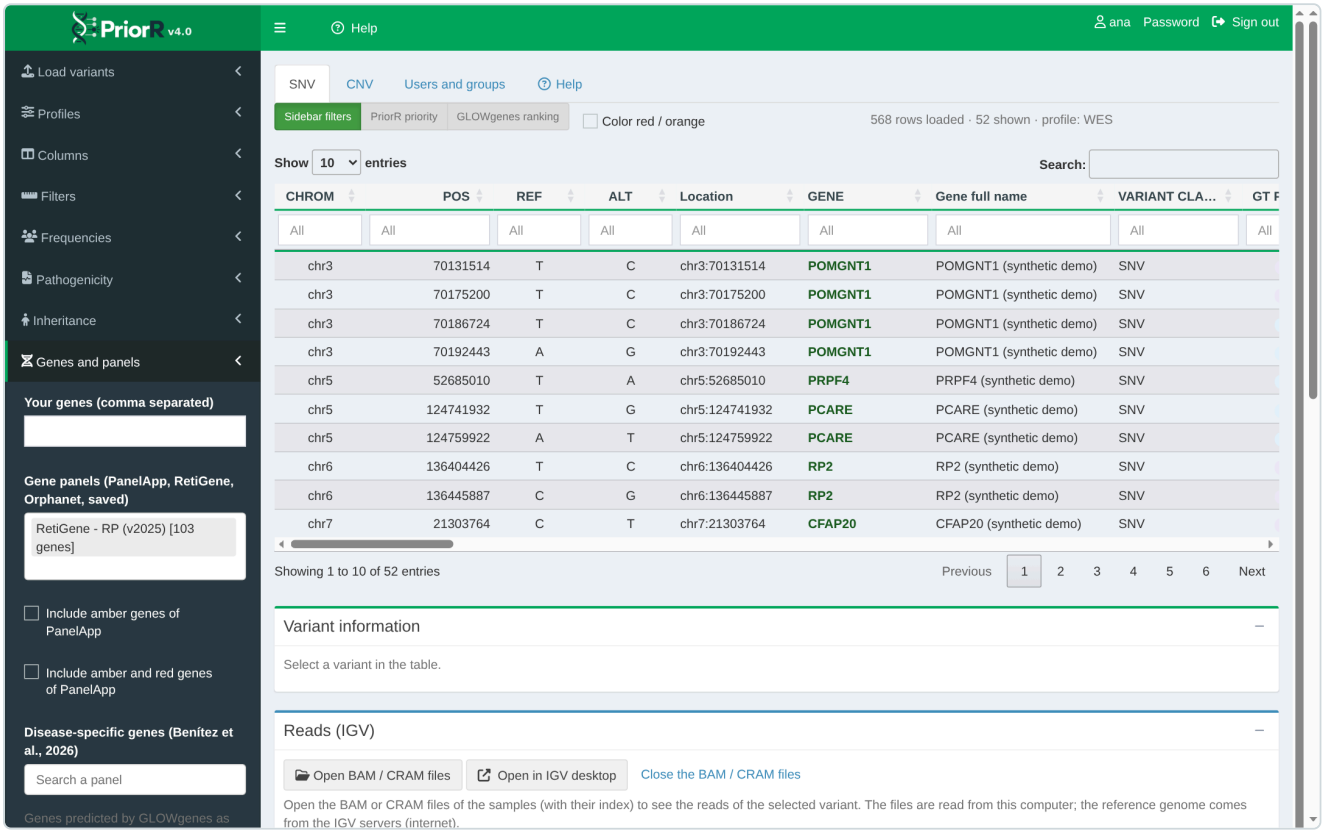


Figure 5. Variants of the RetiGene RP panel.

Phenotypes / HPO (fig. 6): choose HPO terms by name or identifier, or paste a list of identifiers (`HP:0000510, HP:0000556`). The genes of each term **and of its subterms** work like one more panel.

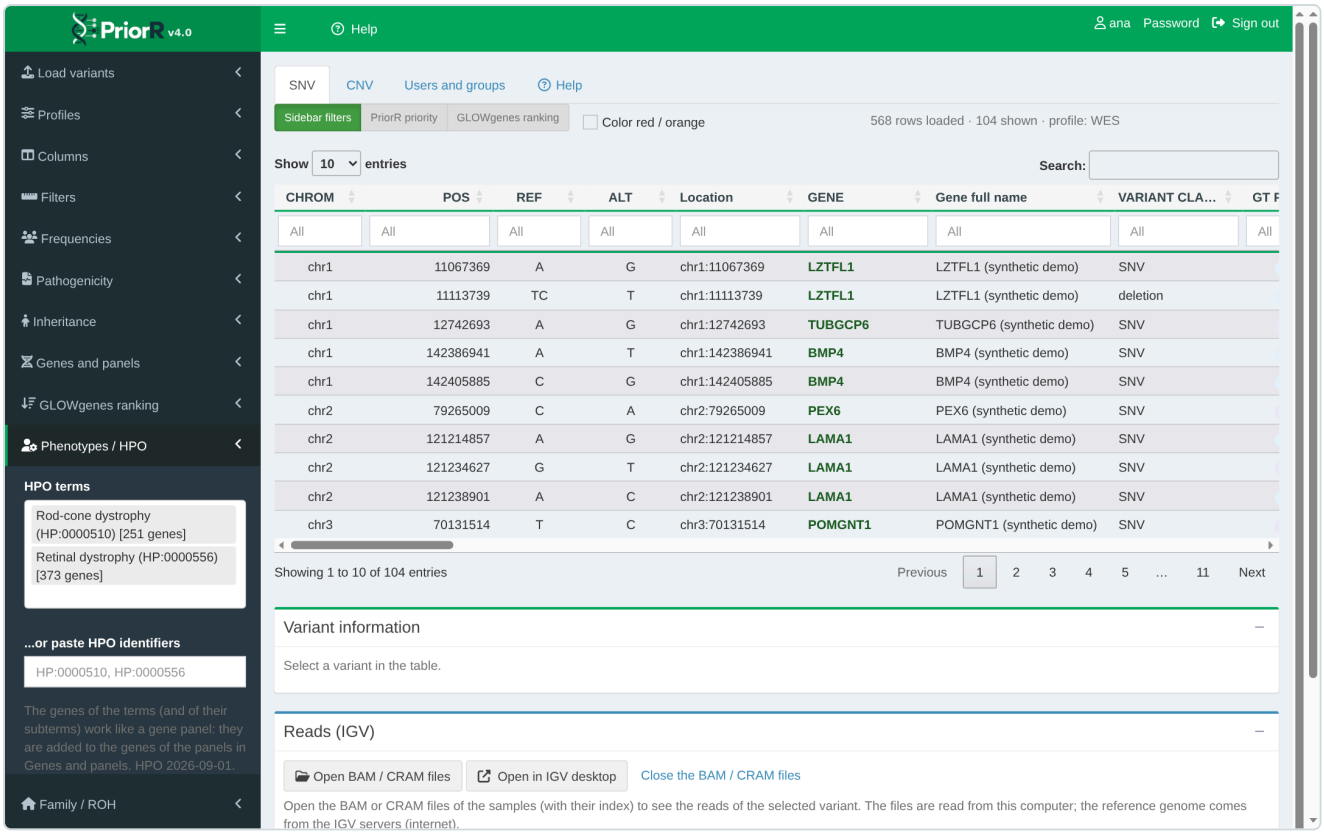


Figure 6. HPO terms as a panel (Rod-cone dystrophy, Retinal dystrophy).

9. Inheritance, trios and de novo variants

In **Family / ROH** choose the family: a **trio** among the samples of the file (proband, sex, father and mother) or a **PED file**. Then choose the pattern in **Inheritance** (fig. 7):

- **De novo:** proband heterozygous (VAF 0.2-0.8) and both parents homozygous reference with depth ≥ 10 and at most one read with the variant. Chromosome X in a male proband: only the mother; Y: only the father. A parent without a call never gives a de novo.
- **Dominant**, **Recessive-Homozygous**, **Recessive-Compound heterozygous** (one variant from each parent when both are available) and **X-linked**, with the OMIM inheritance of the gene.
- Without a family the genotype of the sample is used. **ROH:** colors the variants in runs of homozygosity.

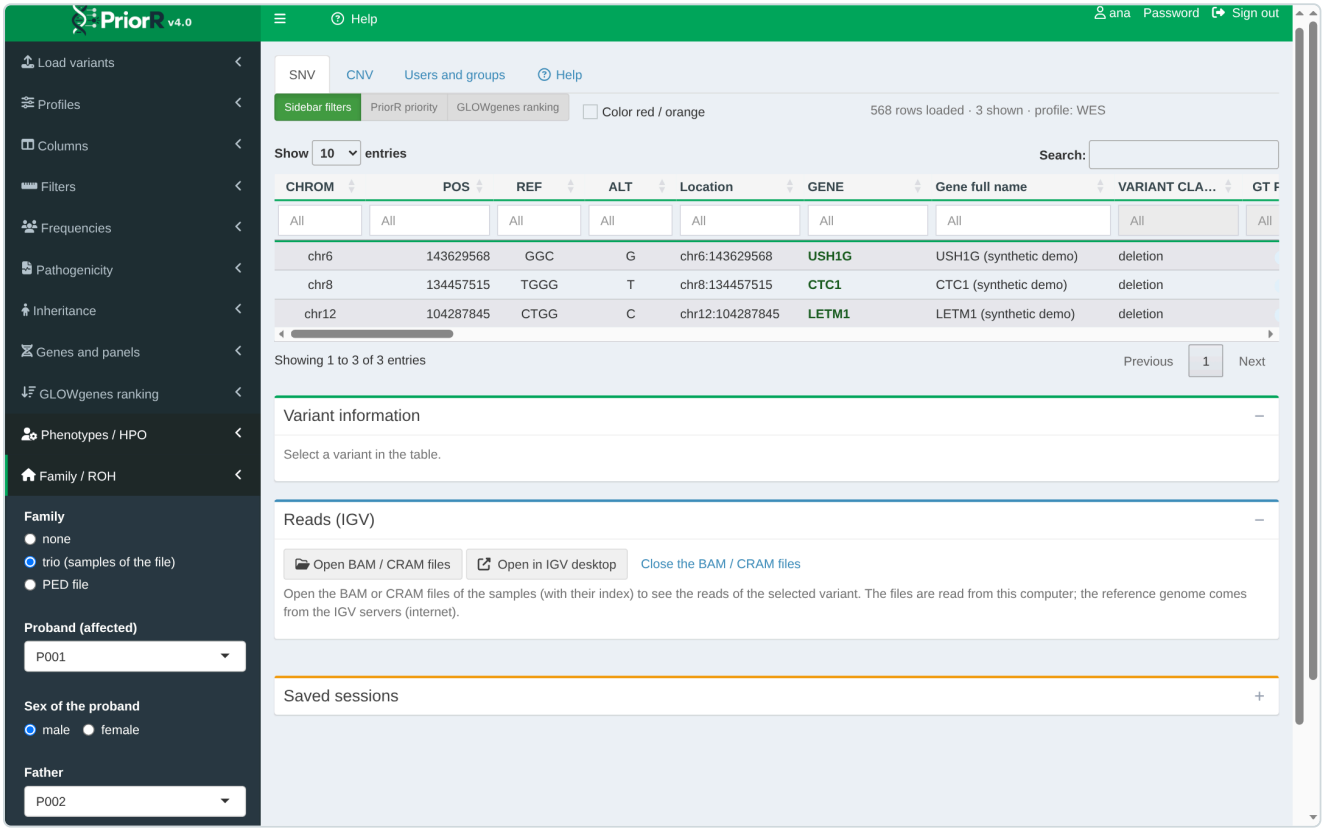


Figure 7. Trio of the demo data and the de novo candidates.

10. Prioritization

Above the table, the **PriorR priority** button shows the variants of the sidebar filters with the red ones at the top, then the orange ones, then the rest. In the *Sidebar filters* view, *Color red / orange* colors them without changing the order.

A box above the table says how many red and orange variants there are and what they mean:

- **Red** (never when ClinVar says benign, likely benign or uncertain significance): ClinVar pathogenic / likely pathogenic; rare (gnomAD genomes popmax < 0.001 or absent) with > 60% pathogenic predictors and CADD > 25 and / or MutScore > 0.8; rare start / stop lost,

stop gained, frameshift or in-frame indel with > 75% pathogenic predictors (without CADD / MutScore); rare splicing or intronic variant with dbscSNV ADA or RF > 0.8.

- **Orange:** low frequency (0.001-0.01) with > 60% pathogenic predictors and CADD > 25 or MutScore > 0.8; the same for splicing / intronic variants with ADA or RF > 0.8; other pathogenic ClinVar annotations (low penetrance, risk factor), not conflicting; rare ClinVar uncertain significance with > 60% pathogenic predictors and CADD > 25 or MutScore > 0.8.

11. GLOWgenes ranking

The **GLOWgenes ranking** button above the table shows only candidate variants (rare, not benign and with at least one evidence of pathogenicity) ordered by the GLOWgenes rank of their gene: 0 are the genes of the disease panel and then the closest genes in the GLOWgenes networks (fig. 8). **Hide panel genes** keeps only the new candidates. In the sidebar, *Rank the genes with*:

- the `GLOWgenes` column of the file;
- **precomputed rankings:** Orphanet panels, 209 PanelApp panels (Benítez et al., Sci Rep 2026) and the rankings computed by the lab;
- **a new ranking:** a panel (10 or more genes) or a gene file is run with GLOWgenes in Docker (image `tblabfjd/glowgenes`, about 15 minutes); the result is shared with the lab.

Only genes up to a GLOWgenes rank limits the rank in both views.

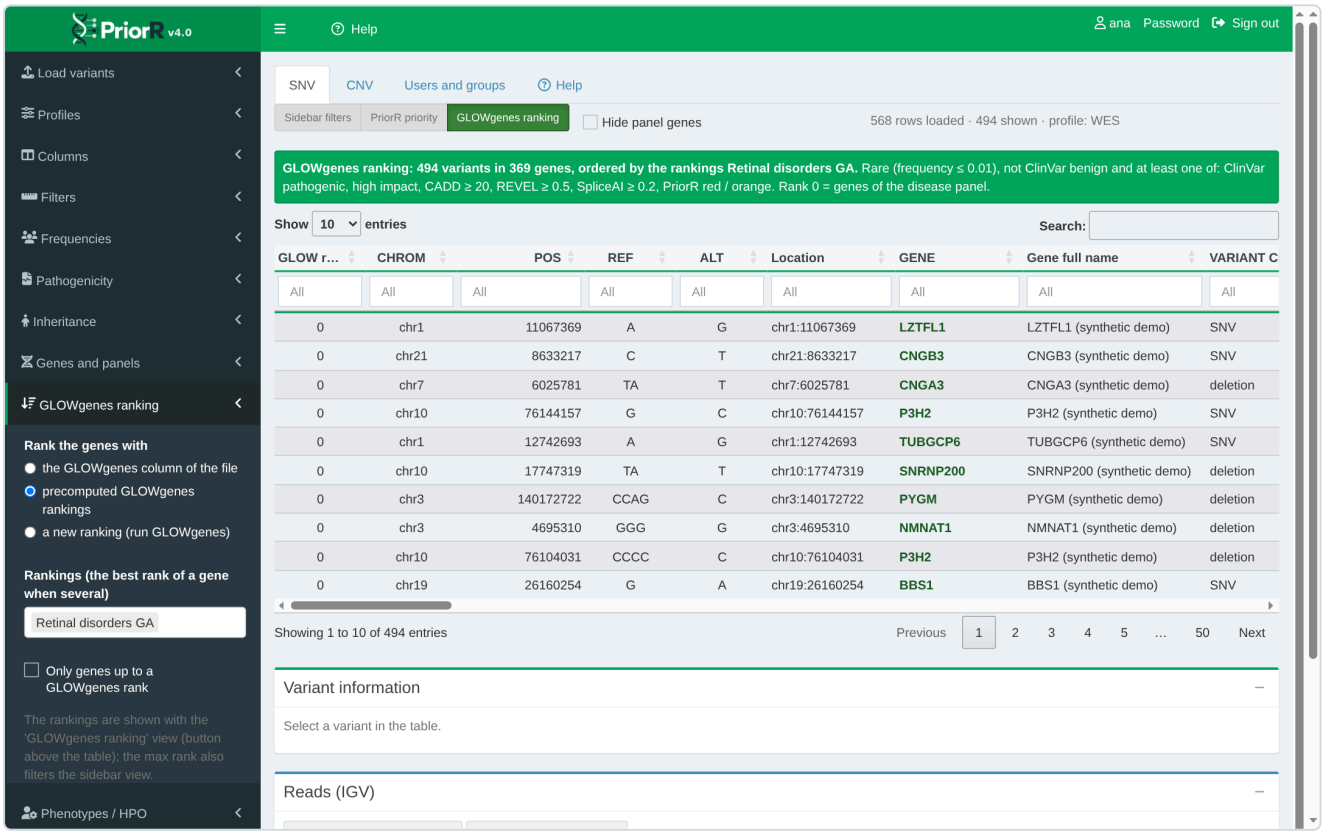


Figure 8. GLOWgenes view with the PanelApp Retinal disorders ranking.

12. Variant information and ACMG criteria

Selecting a row shows (fig. 9): gene, position, impact and ClinVar badges; transcript and HGVS; genotypes of each sample (GT, DP, AD, VAF and variant callers); read depth compared with the whole file; population frequencies; predictors and scores (with AlphaMissense and the gene constraint); OMIM; external links (GeneCards, STRING, OMIM, ClinVar, gnomAD, PanelApp, DECIPHER, GTEx, UniProt, VarSome, SpliceAI Lookup, UCSC...); ACMG criteria and comments.

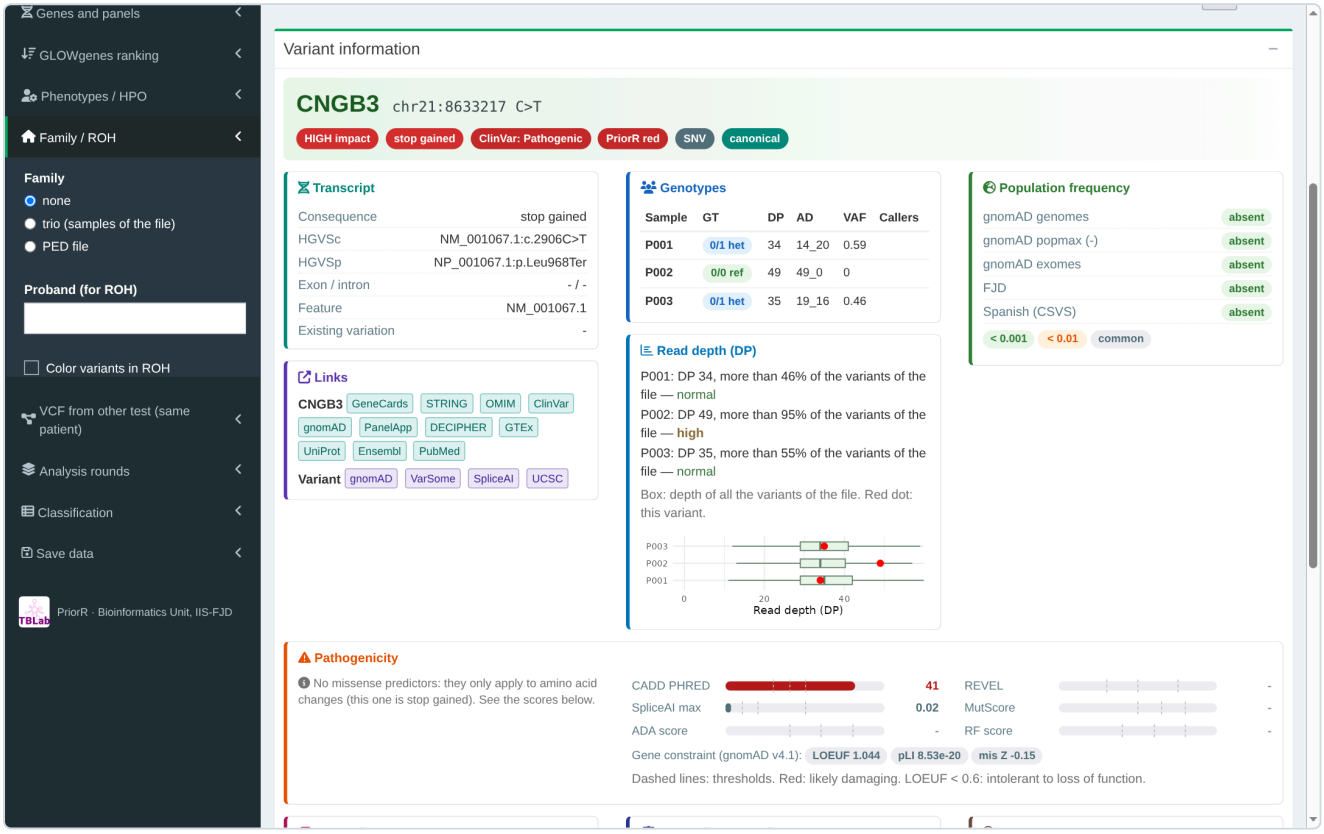


Figure 9. Variant information of a stop variant of CNGB3 in the demo trio.

Gene structure (fig. 10) draws the transcript of the row from 5′ to 3′: exons at scale (coding part thicker, UTRs thinner), introns shortened, the variant in red and its exon or intron in orange, with the distance to its borders in bp (useful for splicing). The exons come from the RefSeq and GENCODE tables of the UCSC Genome Browser, downloaded once for the whole lab into the shared folder (~30 MB, started automatically the first time); then it works without internet. Until then, a schematic drawing with the EXON / INTRON columns.

Gene in panels lists the panels of the library with the gene (PanelApp genes in green, amber or red), whether it is in the genes of the chosen HPO terms, and GLOWgenes and SGDS. **Other variants of the gene in this sample** lists the other variants of the gene that pass the filters with their genotypes; with a family (trio or PED) it says from which parent each one comes and warns of compound heterozygotes in trans (fig. 11).

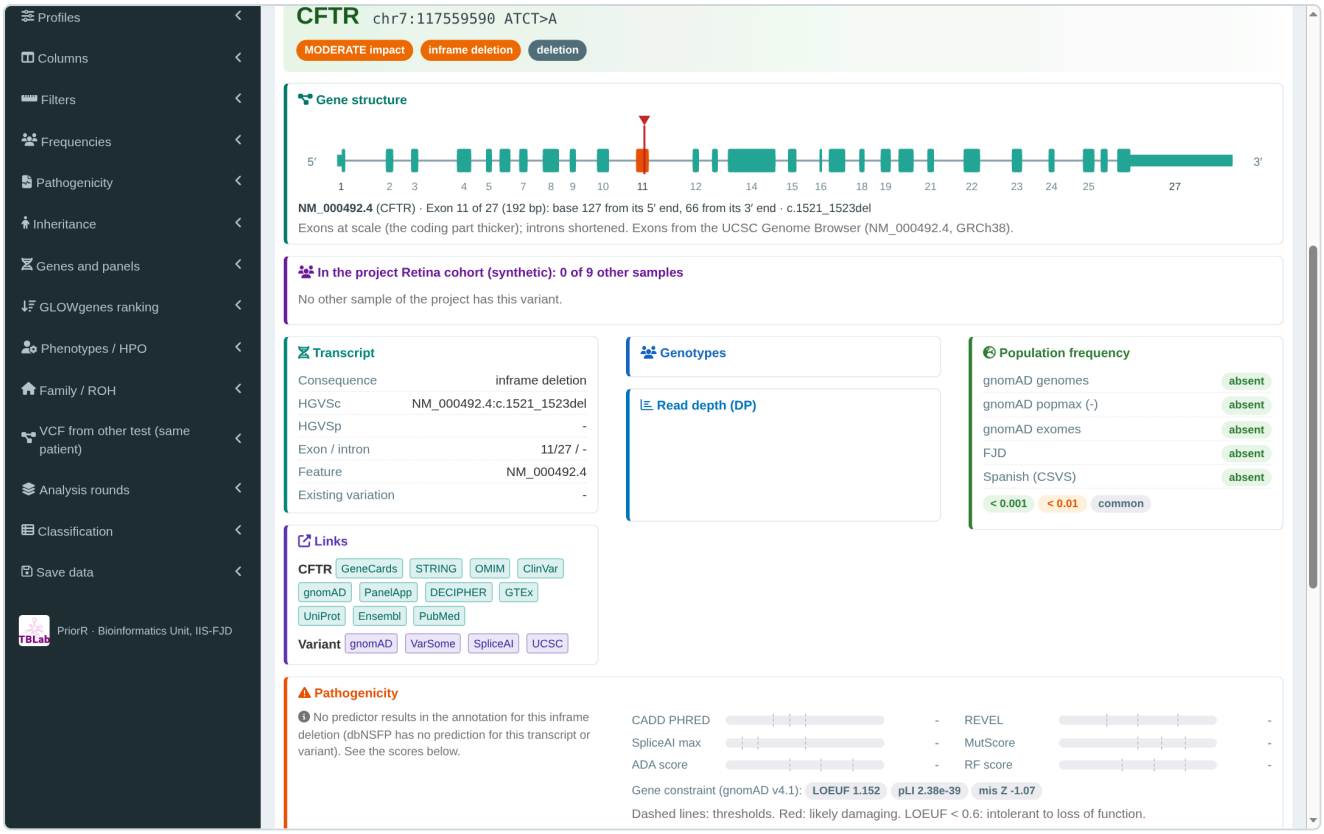


Figure 10. Gene structure of the public CFTR F508del variant (exon 11 of 27), in a small synthetic table.

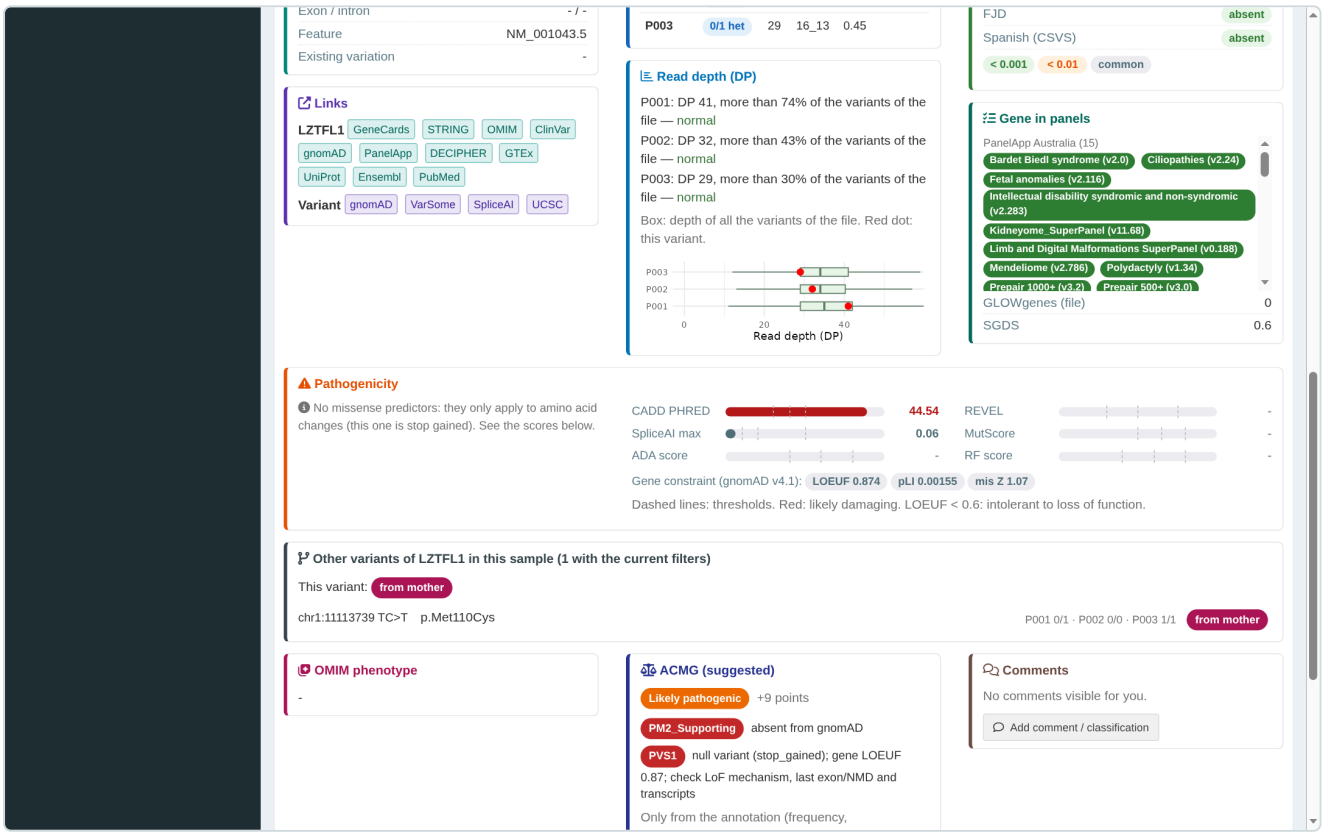


Figure 11. Gene in panels and the other variant of LZTFL1 in the demo trio, with its parental origin.

The **ACMG (suggested)** card (fig. 12) lists the criteria that the annotation supports (ACMG/AMP 2015 with the ClinGen SVI recommendations) with their points and class (Tavtigian et al. 2020): PVS1 (null variant), PM2_Supporting (absent or very rare), PM4, PP3 /

BP4 (REVEL thresholds of Pejaver et al. 2022, CADD without REVEL, SpliceAI), BP7, BA1 / BS1 and PM6 (de novo in the trio), PS1 / PM5 (same amino acid change / another missense at the same residue pathogenic in ClinVar, same transcript). They are suggestions: segregation, functional data, phenotype or *in trans* are added by the analyst.

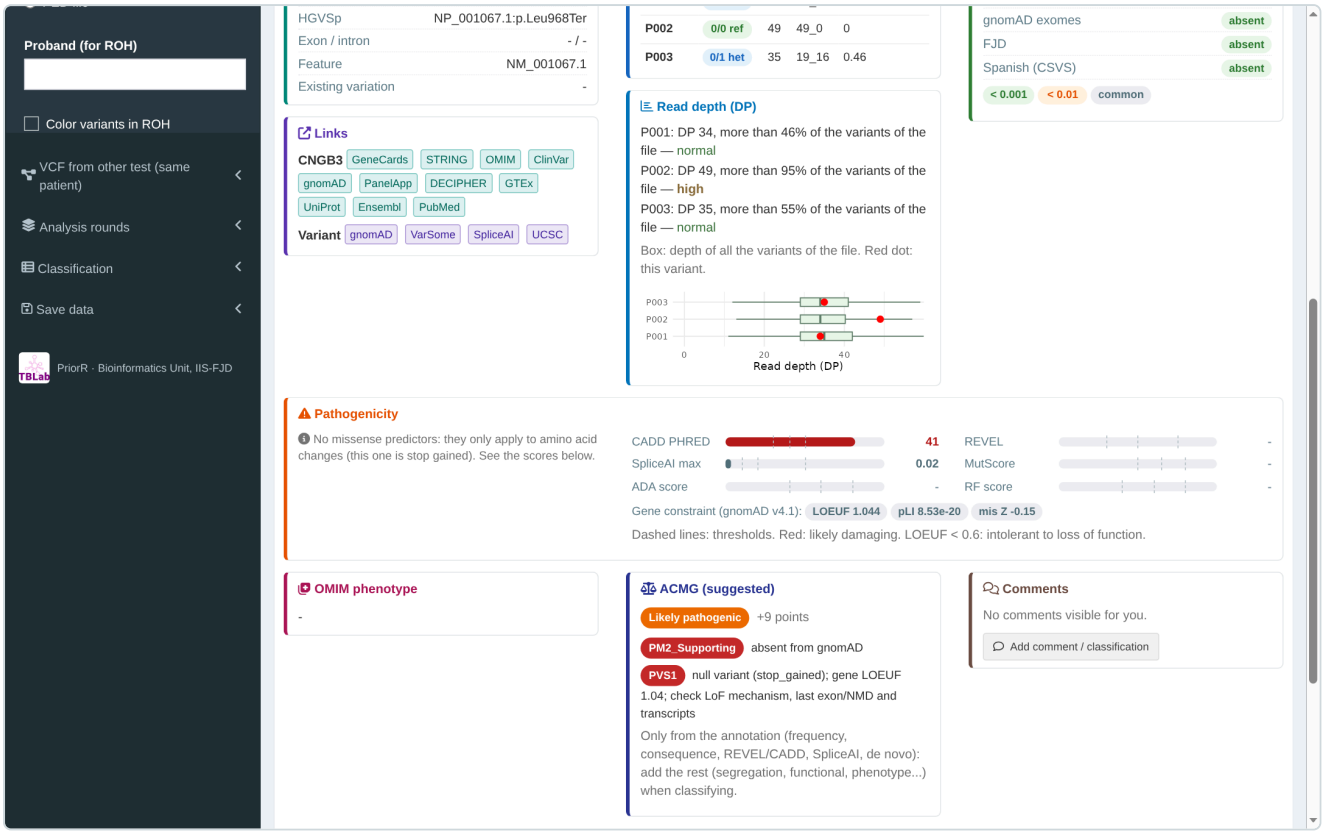


Figure 12. Predictors, gene constraint and the ACMG criteria suggested from the annotation.

13. Classifications, comments and reanalysis

Add comment / classification (fig. 13): classification (Benign ... Pathogenic), ACMG criteria (the suggested ones come chosen: remove, change or add, and the points and class are recalculated), comment and who sees it: *all my groups*, one group or only me. Classifications belong to the variant (chromosome, position, alleles), so they are seen in every file that has it. **Classification > Show classifications and comments** adds them to the table.

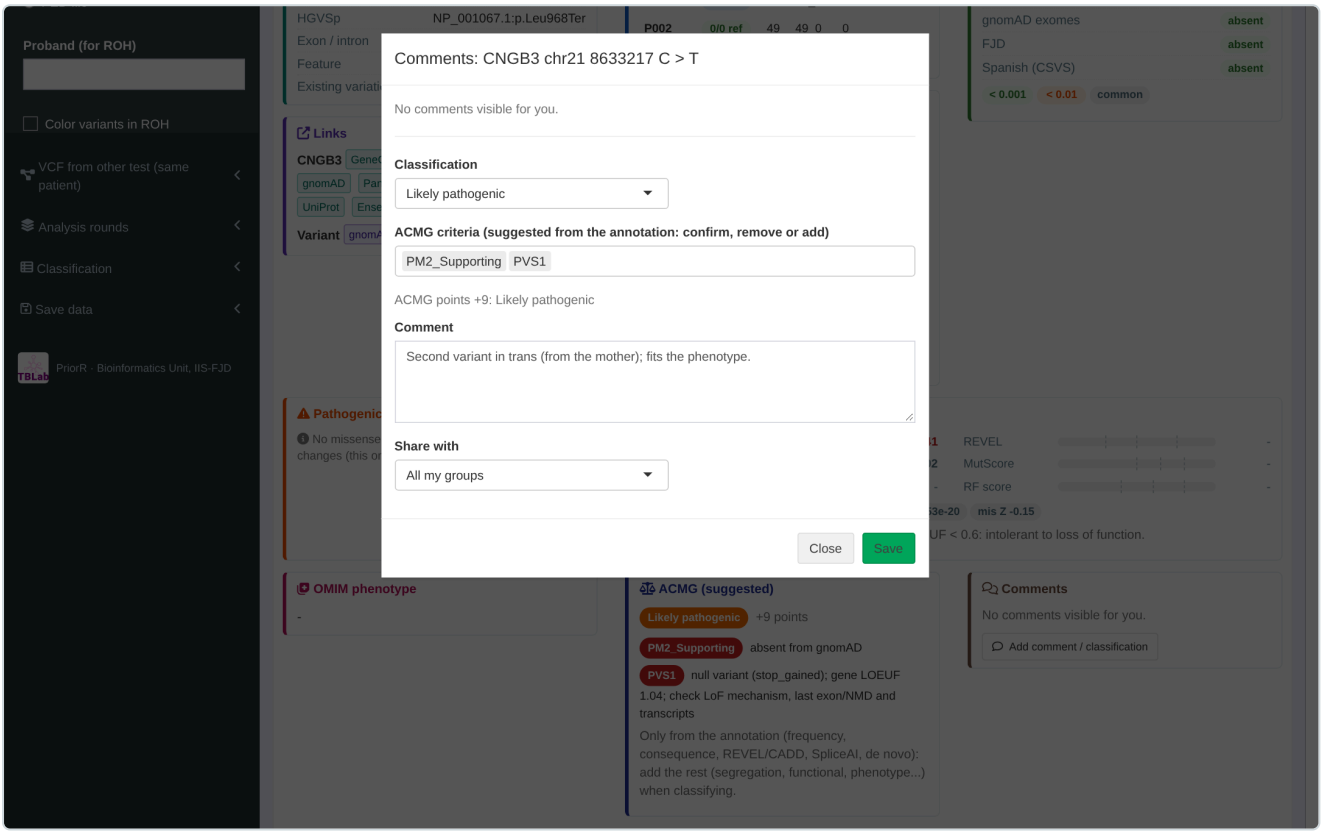


Figure 13. Classifying a variant with its ACMG criteria.

Reanalysis. When a variant is classified, PriorR saves what the annotation said (ClinVar, gnomAD popmax, green PanelApp panels of the gene, ACMG suggestion). When a newer annotation of the patient is opened, a box lists the classified variants that changed (fig. 14); *Only classified variants with changes* shows only them.

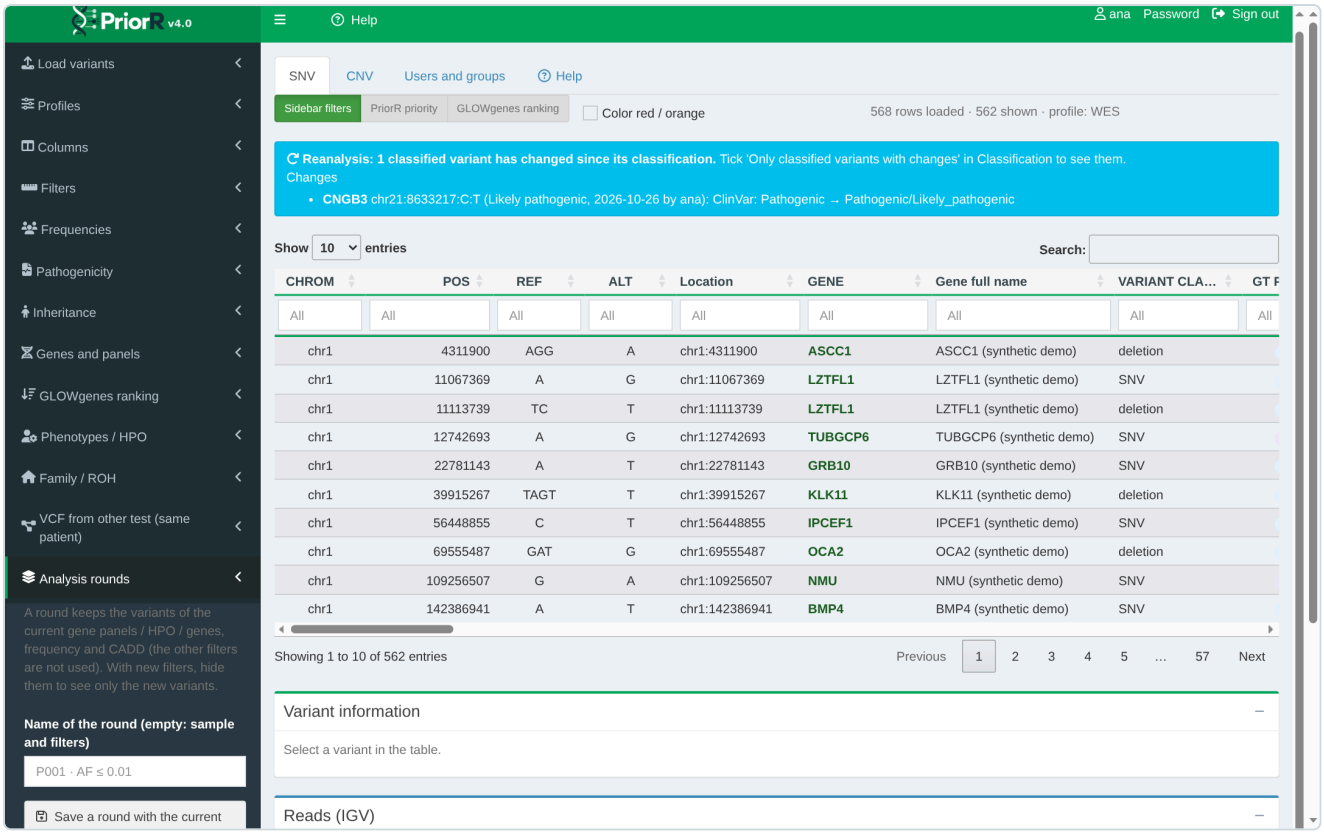


Figure 14. Reanalysis: the ClinVar classification of a classified variant changed.

14. Analysis rounds

To analyze a genome in steps without reviewing the same variants again: in **Analysis rounds**, *Save a round with the current panels, frequency and CADD*. A round keeps the variants of the current gene panels / HPO / genes, maximum frequency and minimum CADD (the other filters are not used) and its default name says so (e.g. `P001 · RetiGene - RP (v2025) · AF ≤ 0.01 · CADD ≥ 20`). With new filters, tick **Hide the variants of previous rounds** to see only new variants (fig. 15). Rounds are saved per user and file and can be deleted.

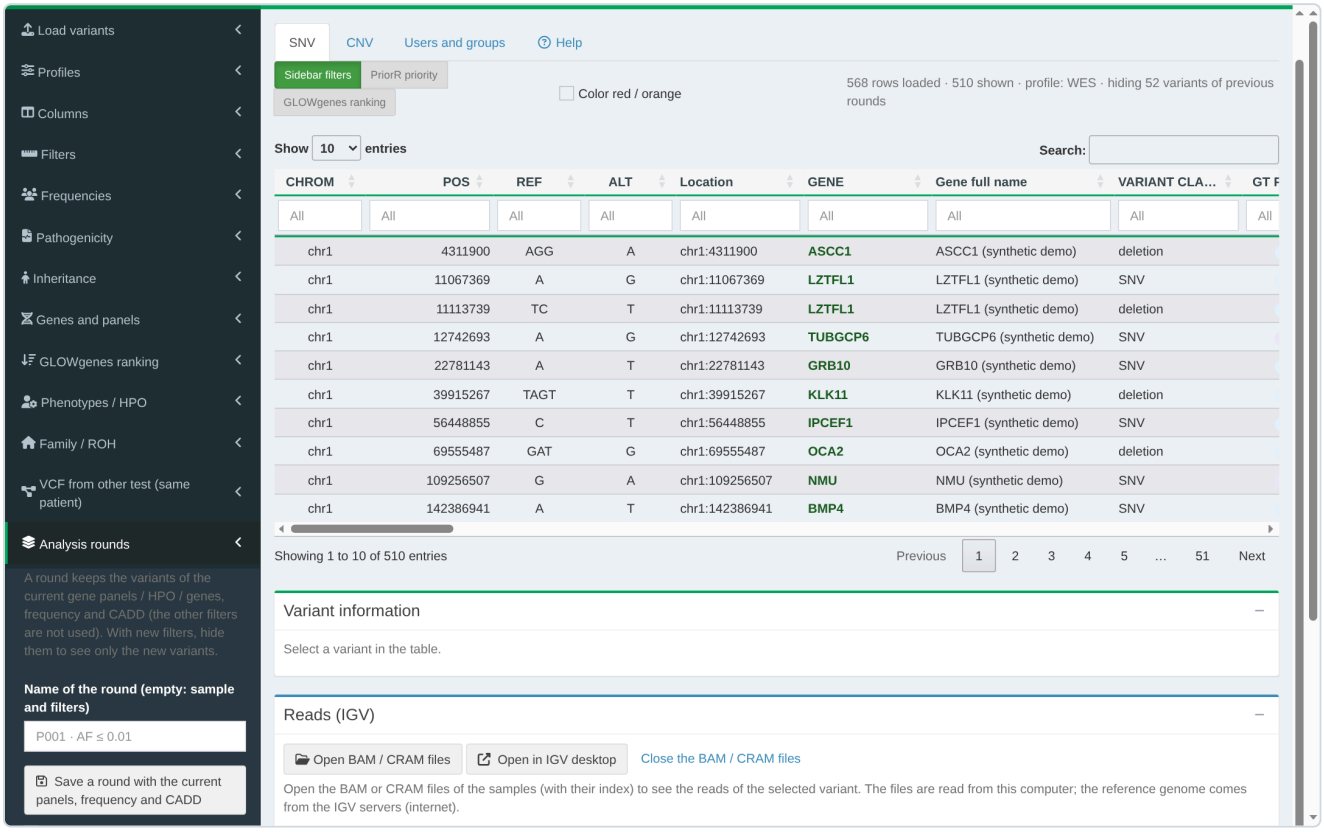


Figure 15. A saved round and its variants hidden in the next analysis.

15. Projects (other samples)

A **project** groups the variant files of several samples, usually one file per sample (a file with several samples, e.g. a trio, counts each sample by its `<sample>_GT` column). In **Project (other samples) > Manage projects** (fig. 16): create a project and add its files. Each file is read once in the background and only its variants and genotypes are kept in the shared folder (a genome takes a few MB); changed files are read again with *Index the new files*.

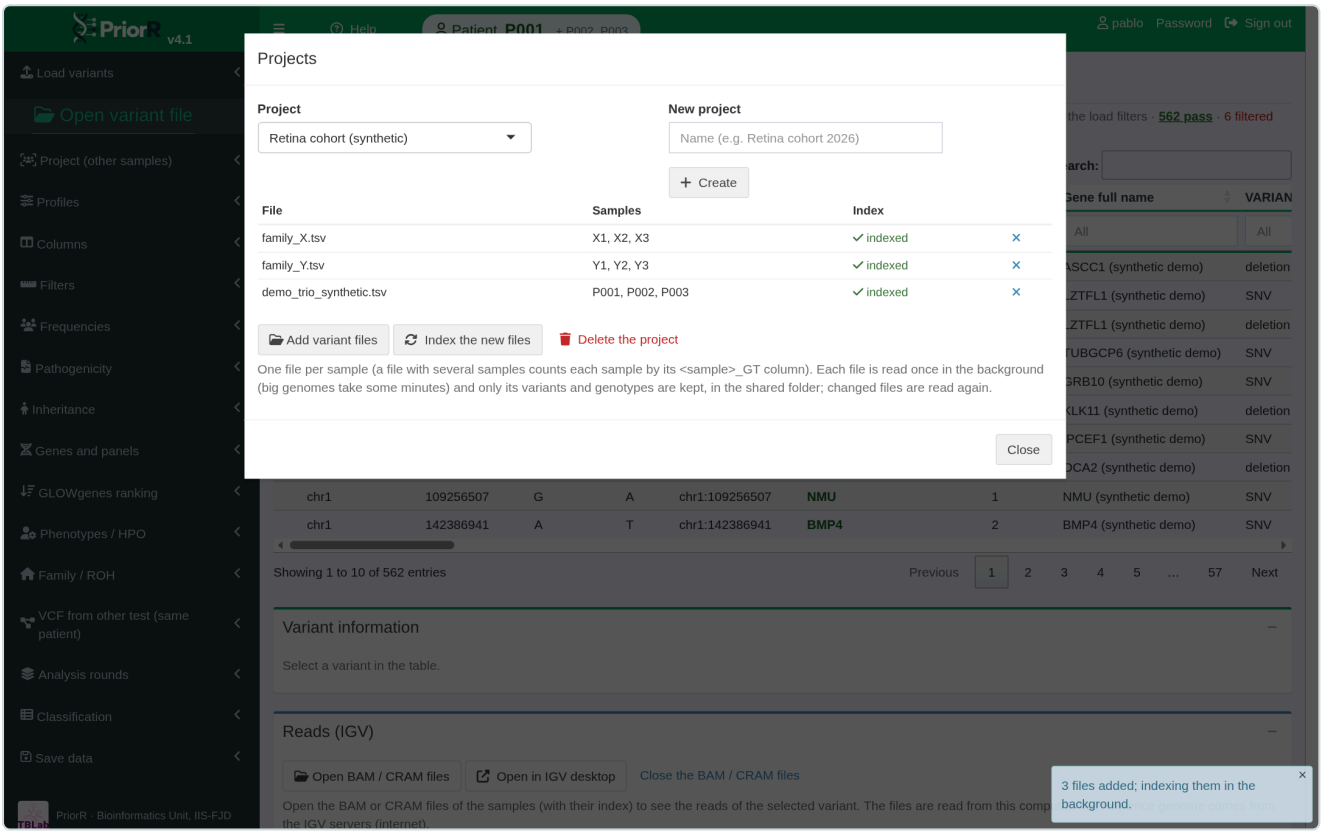


Figure 16. A project of three synthetic families, indexed.

To analyse the samples of a project one after another, choose the project and a sample in *Open a sample of the project* and click **Open**: the *Open variant file* dialog opens with its file chosen. With the project chosen in **Project**, the table gets **In project** (other samples with the variant) and **Hom in project** (of them, homozygous) next to GENE; the samples of the loaded file are not counted. *Hide variants in more than this number of other samples* removes recurrent artefacts or variants frequent in the population. *Variant information* lists the other samples with the variant, their genotype and file (fig. 17).

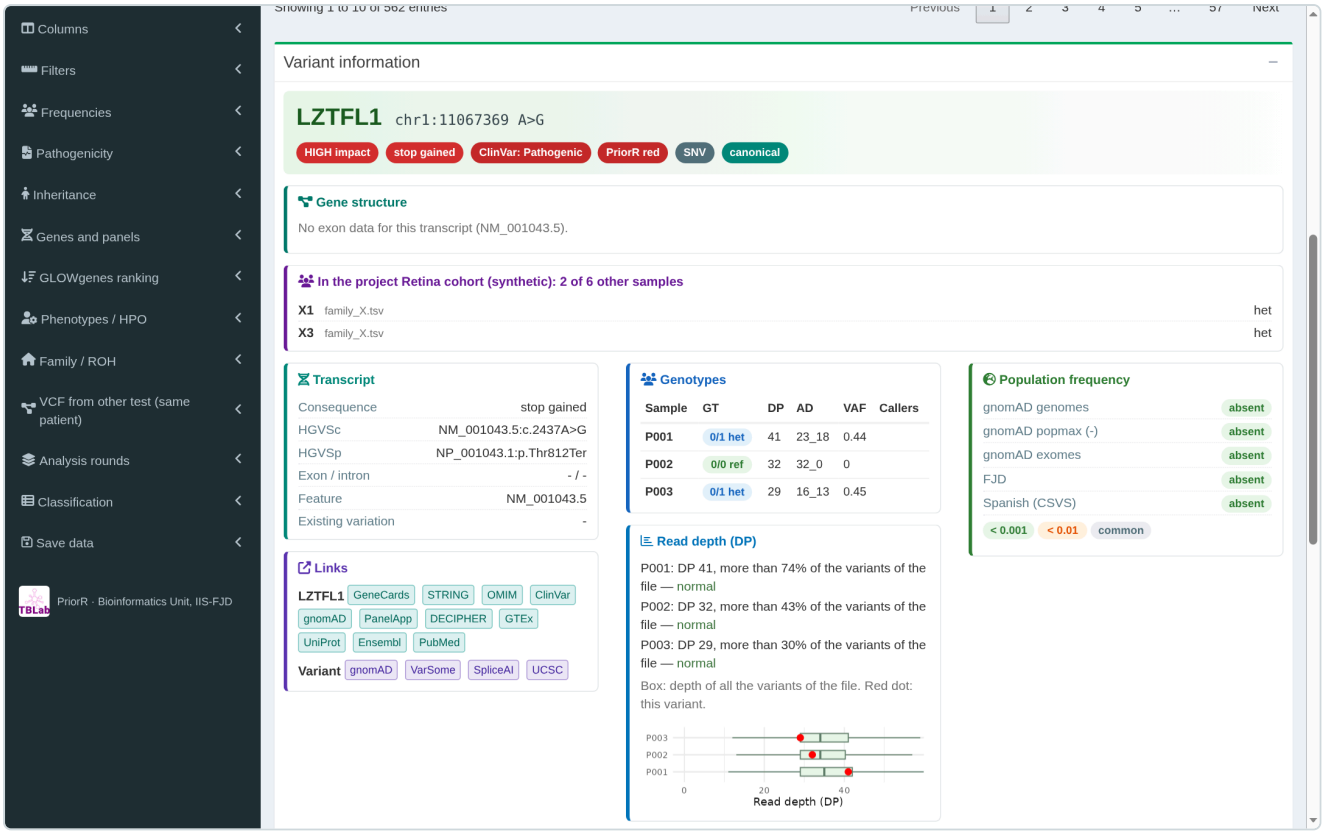


Figure 17. The other samples of the project with the selected variant.

16. Reads (BAM / CRAM)

The **Reads (IGV)** box shows the reads of the selected variant with igv.js: *Open BAM / CRAM files* (with their index next to them: `X.bam.bai`, `X.bai`, `.csi`, `.crai`, also in capitals; each file is assigned to the sample whose name it contains). The files are read from the computer; the reference genome comes from the IGV servers. *Open in IGV desktop* sends the files and the position to IGV (port 60151 enabled). In CNVs, small CNVs are shown whole and big ones by their two breakpoints.

17. CNVs

The **CNV** tab opens AnnotSV files (fig. 18). Filters: samples, annotation mode (*full*: one row per CNV; *split*: one per gene), type, location, minimum ACMG class, number of programs, genes, panels, disease-specific genes, HPO and frequency of benign gains / losses. **GLOWgenes ranking** gives each CNV the best rank of its genes (`GLOW_rank`), to order or limit. The CNV information shows size, class, genes, known pathogenic CNVs, OMIM / GenCC, constraint and links (fig. 19). As in the SNVs, the header shows the patient (the sample chosen in *Samples*), the line above the table counts the CNVs that **pass** the filters and those **filtered out** (of the chosen samples and annotation mode; click to see them) and the columns can be moved by dragging their header. **Gene structure** draws in one line the region of the CNV at genomic scale with its genes and exons, the CNV shaded (red for deletions, blue for duplications) and its exons in that color, and says which exons of each gene it takes (fig. 20).

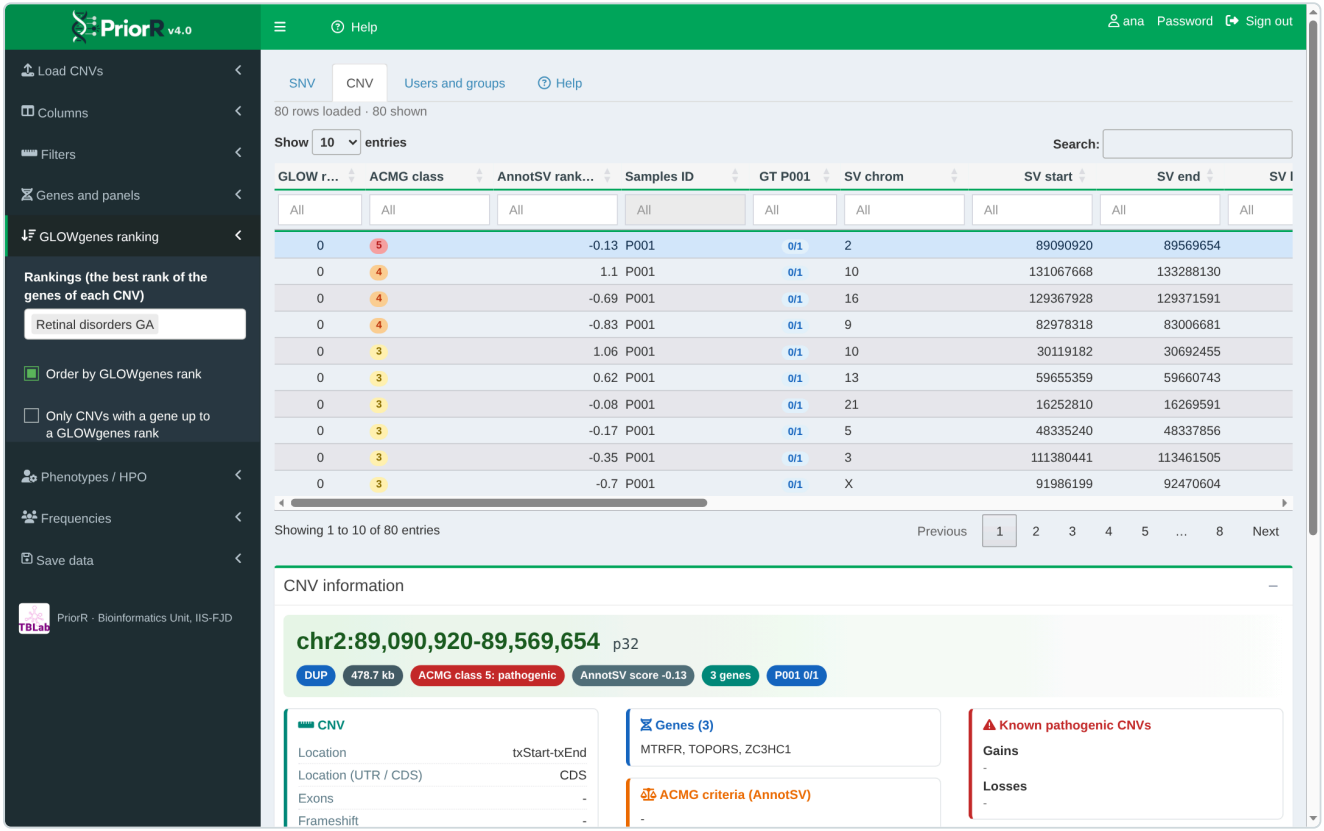


Figure 18. CNVs of the demo ordered by the GLOWgenes rank of their genes.

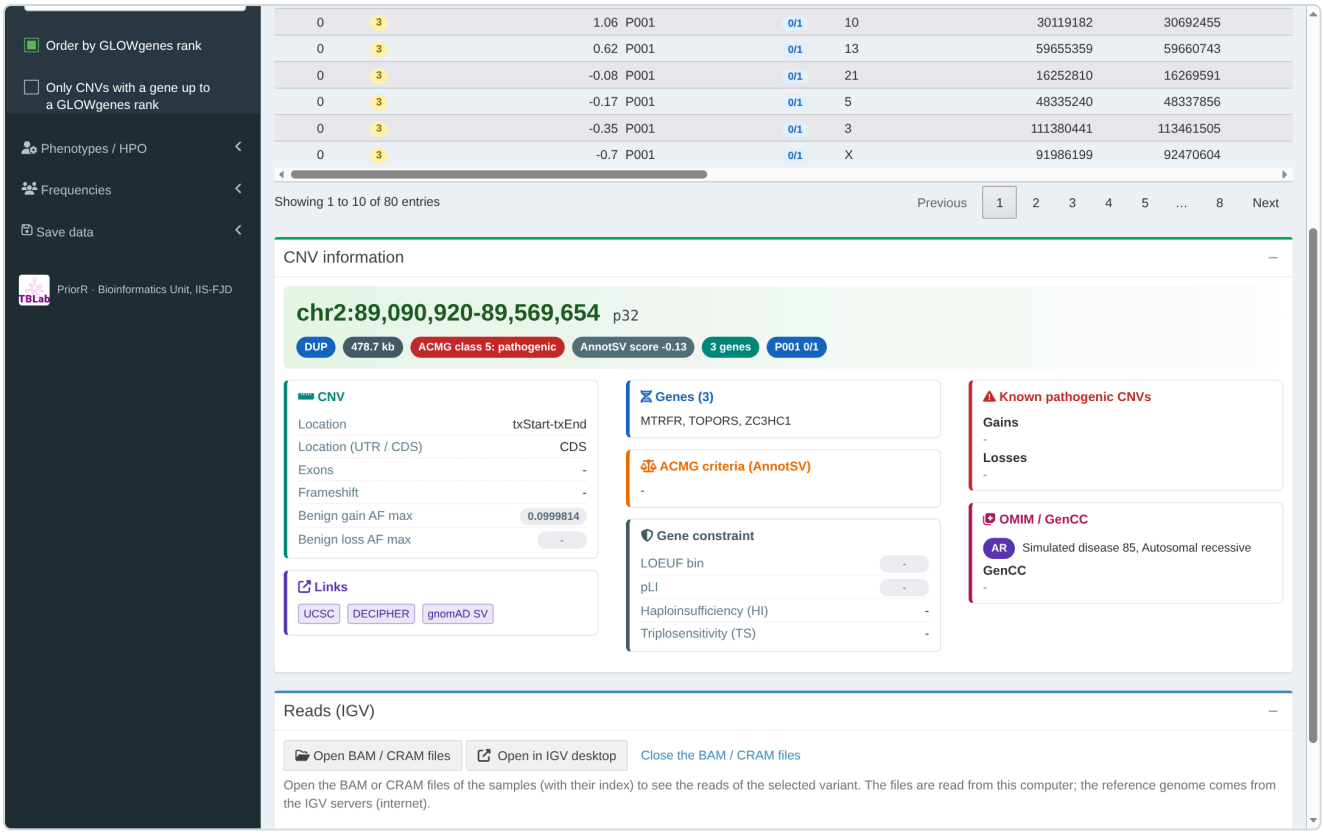


Figure 19. CNV information.

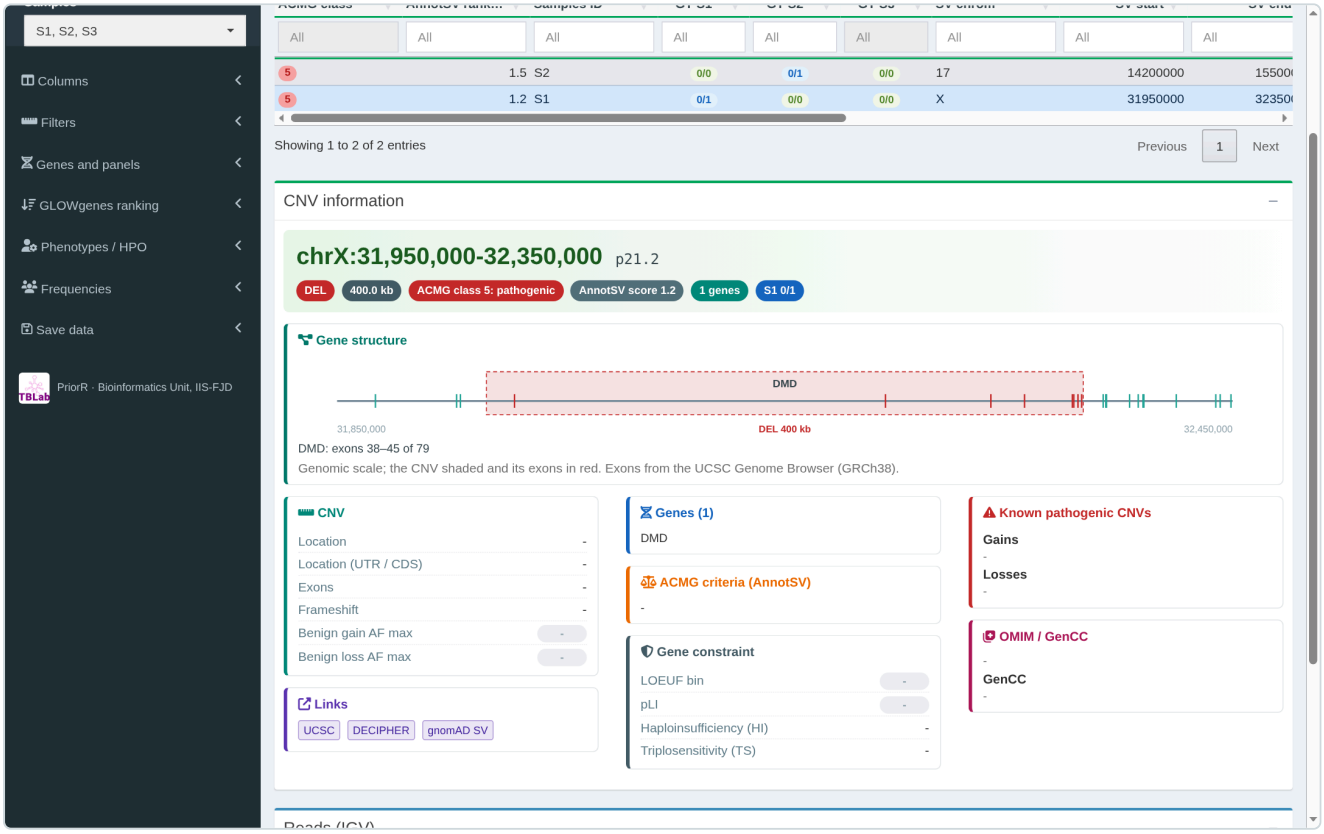


Figure 20. Gene structure of a deletion of DMD exons 38–45 (synthetic multi-sample CNV table with public gene coordinates).

18. Users, groups and sessions

Users and groups (administrators, fig. 21): create users, change passwords, deactivate users, make administrators, create groups and assign members. Members of a group see the comments shared with their groups. From the command line, `Rscript Program_files/scripts/priorr_admin.R reset-password <user>` resets a password. **Saved sessions** keeps an analysis (file, profile, filters, columns, BAM files) to open it again later, also from another computer.

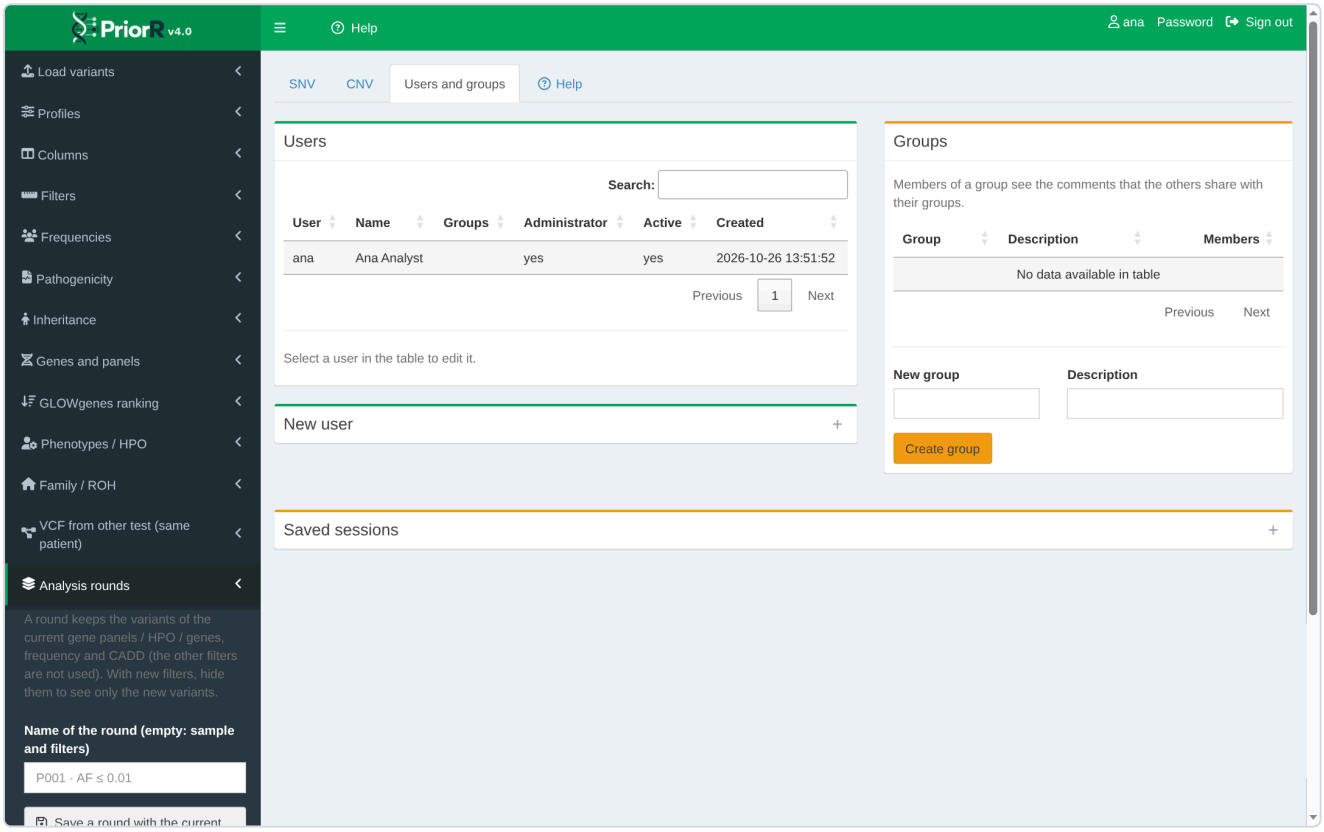


Figure 21. Users and groups.

The **Help** tab (fig. 22) has the complete documentation and the configuration of the computer.

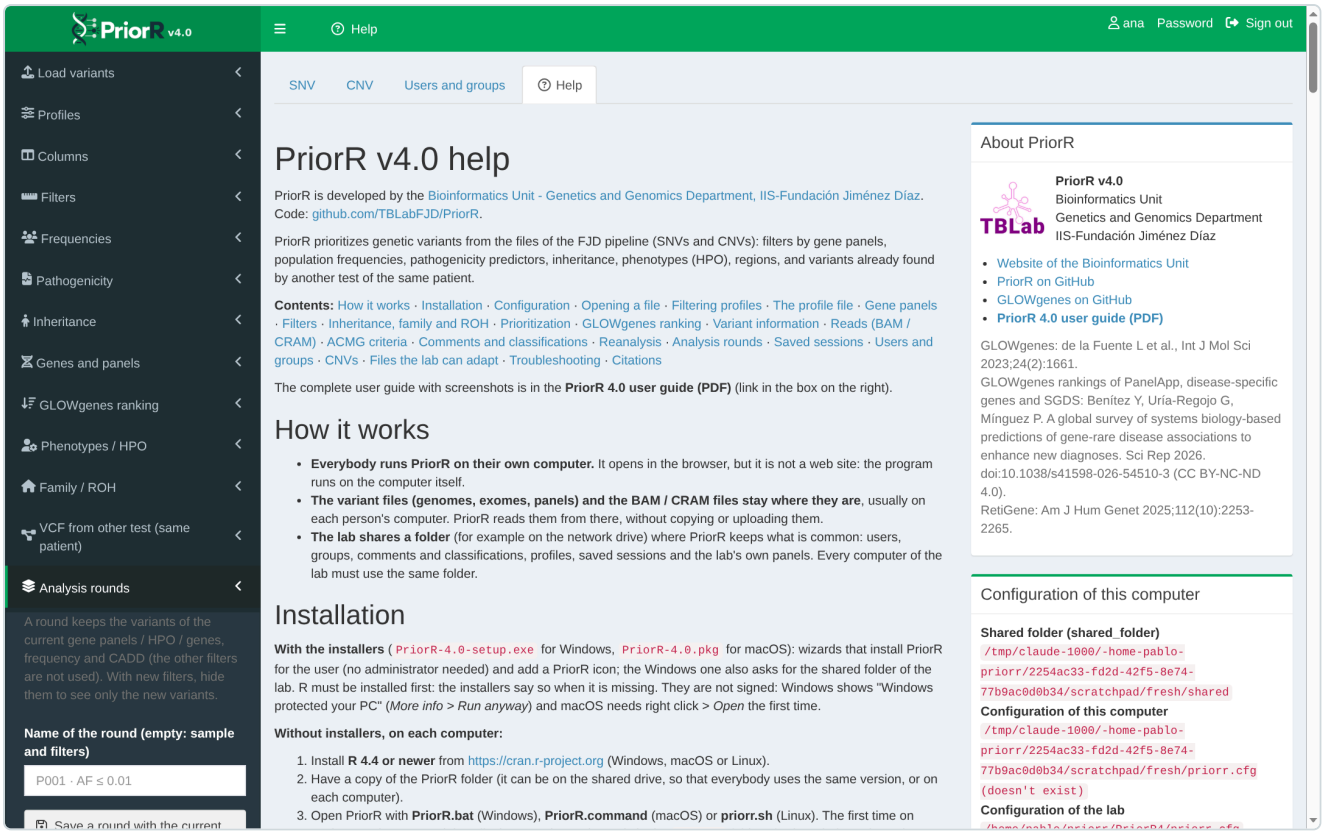


Figure 22. In-app Help.

19. Updating the resources

From the `app/` folder (they need internet):

Resource	Command
PanelApp (lab copy)	`Rscript Program_files/scripts/update_panels.R`
HPO	`Rscript Program_files/scripts/update_hpo.R`
gnomAD gene constraint	`Rscript Program_files/scripts/update_constraint.R`
AlphaMissense (once, ~640 MB)	`Rscript Program_files/scripts/get_alphamissense.R`
ClinVar missense (PS1 / PM5)	`Rscript Program_files/scripts/update_clinvar_missense.R`
RetiGene	`Rscript Program_files/scripts/import_retigene.R`

Other files the lab can adapt: `Program_files/profiles/*.json` (built-in profiles), `links.tsv` (external links), `glowgenes.json` (filters of the GLOWgenes view).

20. Troubleshooting

- **The address of PriorR:** `http://127.0.0.1:8484`; when a second PriorR is open, the next port (8485...).
- **A project counts nothing:** check in *Manage projects* that its files are *indexed* (not pending or not found).
- **No gene structure drawing:** the transcript (Feature) must be a RefSeq (NM_, NR_, XM_, XR_) or Ensembl ENST one; the first time the table of transcripts is downloaded (internet,

a minute). By hand: `Rscript Program_files/scripts/get_transcripts.R` from `app/`.

- **The first load of a genome is slow:** it prepares a cache; the next loads take seconds.
- **I don't see the comments of a colleague:** check that you share a group and that the comment is not private.
- **Each computer sees different users:** they don't use the same shared folder (see Help > configuration).
- **The reads don't show:** index next to the BAM / CRAM, internet for the reference, `genome_build` of the files.
- **A BAM doesn't appear in the dialog:** check that the `.bam` is really in the folder (sometimes only the `.bai` / `.md5` were copied).
- **"Permission denied" in a network folder:** folders mounted with sshfs only let in the processes of the user who mounted them; start PriorR from that user's session.
- **Run GLOWgenes says Docker is not available:** install Docker, start it and (Linux) add the user to the `docker` group.

21. Citations

- GLOWgenes: de la Fuente L, Del Pozo-Valero M, Perea-Romero I, Blanco-Kelly F, Fernández-Caballero L, Cortón M, Ayuso C, Mínguez P. Prioritization of new candidate genes for rare genetic diseases by a disease-aware evaluation of heterogeneous molecular networks. Int J Mol Sci 2023;24(2):1661.
- GLOWgenes rankings of PanelApp panels, disease-specific genes and SGDS: Benítez Y, Uría-Regojo G, Mínguez P. A global survey of systems biology-based predictions of gene-rare disease associations to enhance new diagnoses. Sci Rep 2026. doi:10.1038/s41598-026-54510-3.
- RetiGene: Am J Hum Genet 2025;112(10):2253-2265. PanelApp (Genomics England, Australia). Human Phenotype Ontology.
- AlphaMissense: Cheng J et al. Science 2023. ClinVar: Landrum MJ et al. Nucleic Acids Res 2018. gnomAD v4.1. ACMG/AMP: Richards S et al. Genet Med 2015; Tavtigian SV et al. Hum Mutat 2020; Pejaver V et al. Am J Hum Genet 2022; Walker LC et al. Am J Hum Genet 2023.
- Exons of the transcripts: RefSeq (NCBI) and GENCODE tables of the UCSC Genome Browser (https://genome.ucsc.edu).
- PriorR is developed by the Bioinformatics Unit - Genetics and Genomics Department, IIS-Fundación Jiménez Díaz: https://www.translationalbioinformaticslab.es · https://github.com/TBLabFJD/PriorR